100 Days of Azure DevOps

Day 11 Handout — 4 Git areas, object models & commit DAG

Learning in public · Personal lab only · Educational content · Not a sales pitch

Architecture A — the 4 areas of git

Area	Location / State	Purpose in DevOps Workflow
Working Tree	Filesystem directory	Unstaged edits, live code changes
Staging Area (Index)	.git/index	Curated snapshot boundary for atomic commits
Local Repo	.git/objects (DAG)	Committed, immutable history points with SHA-1s
Remote Repo	Azure Repos	Shared upstream collaboration hub triggering CI runs

Architecture B — git object model & graph (DAG)

Git Primitive	Internal Structure	Why it matters in CI/CD
Blob	Compressed file contents hashed by SHA	Deduplication across commits & branches
Tree	List of SHA pointers to blobs and subtrees	Captures exact directory structure snapshots
Commit	Root tree SHA + Author + Committer + Parent SHA	Creates immutable, traceable history graph
Branch / Ref	41-byte text pointer in .git/refs/heads/	Cheap, instant isolation for feature branches

One-liner to remember

Git doesn't store file diffs — it stores snapshot DAGs · Treat your commit history like production code, not a scratchpad

Step-by-step lab (20–30 min)

In your local clone of **azure-100-labs**:

1. Verify status and branch: **git status** and **git branch**
2. Create & switch to feature branch: **git switch -c feature/day11-git-notes**
3. Create notes file: **notes/day-11-git-model.md** with 4-areas summary
4. Stage file explicitly: **git add notes/day-11-git-model.md**
5. Commit with clear intent: **git commit -m 'docs: add Day 11 git mental model'**
6. Inspect commit graph: **git log --oneline --graph --decorate -n 5**
7. Push to remote with upstream tracking: **git push -u origin feature/day11-git-notes**

Core Git CLI commands

```
git switch -c feature/day11-git-notes
git add notes/day-11-git-model.md
git commit -m 'docs: add Day 11 git notes'
git push -u origin feature/day11-git-notes
git log --oneline --graph --decorate -n 5
```

Done checklist

- I can explain the 4 Git areas and Blob/Tree/Commit objects
- Created and switched branches using git switch
- Created an atomic commit and inspected the graph
- Pushed feature branch to Azure Repos with -u tracking

Tomorrow — Day 12

Branching Strategies for CI/CD — Trunk-based development vs GitFlow & Release branches.

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